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CellQuantX Report

Cell Detection & Counting from Microscopy Images

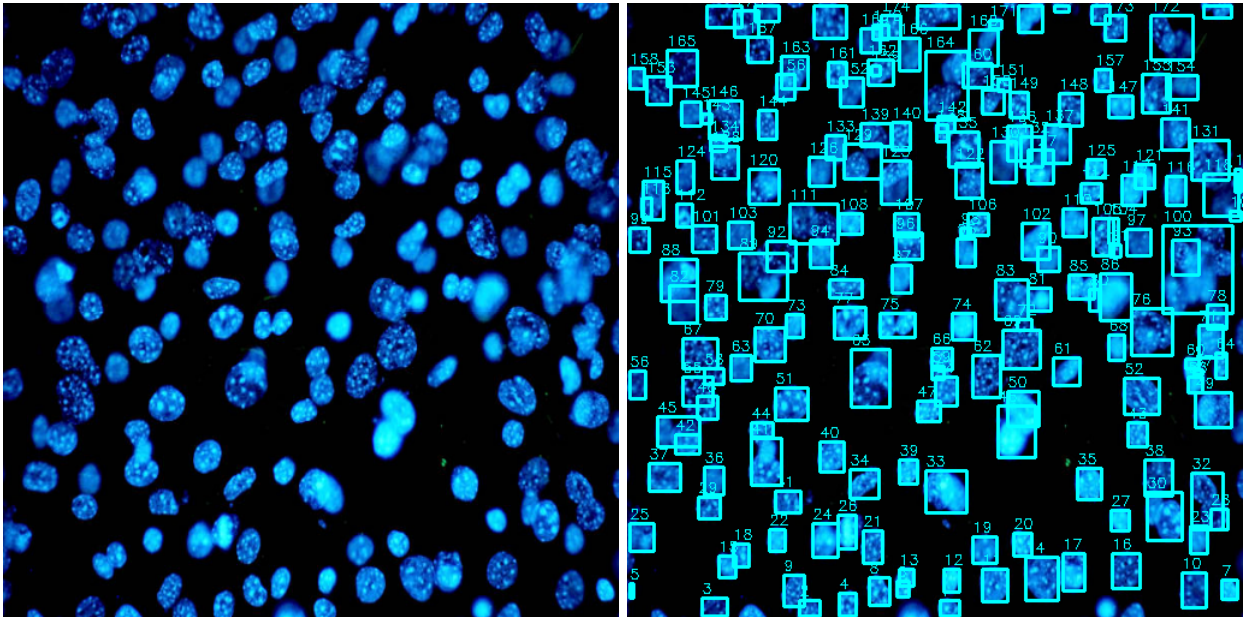
Summary Metrics

Metric	Value
Image	gettyimages-173543371-612x612.jpg
Cell Type	Unknown
Segmentation Method	Global Threshold
Calibration (µm/px)	0.500
Total Cells Detected	183
Cell Density (cells/mm²)	2792.36
Confluence / Coverage	22.1%
Mean Cell Area (µm²)	71.55
Mean Circularity	0.6547
Mean Aspect Ratio	0.8961
Mean Solidity	0.8678
Mean Intensity (ADU)	203.09

AI Biological Interpretation

Automated analysis identified 183 cells in the field with high density and a mean area of 71.5 µm². Mean circularity of 0.655 suggests moderately circular morphology, which is normal physiological conditions. This indicates the sample is well-segmented and may require further review if the morphology appears abnormal.

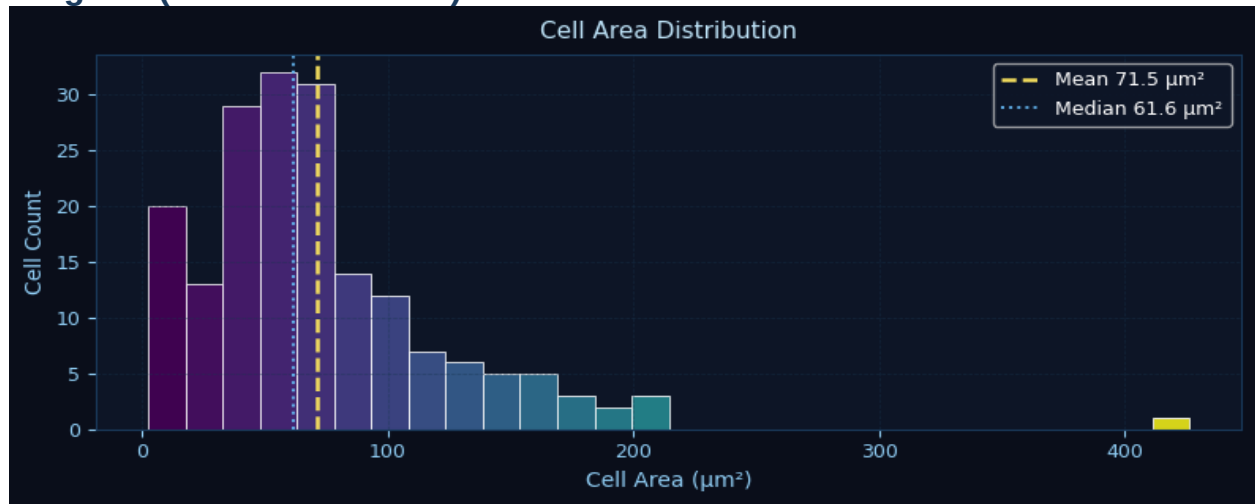
Image Analysis Results



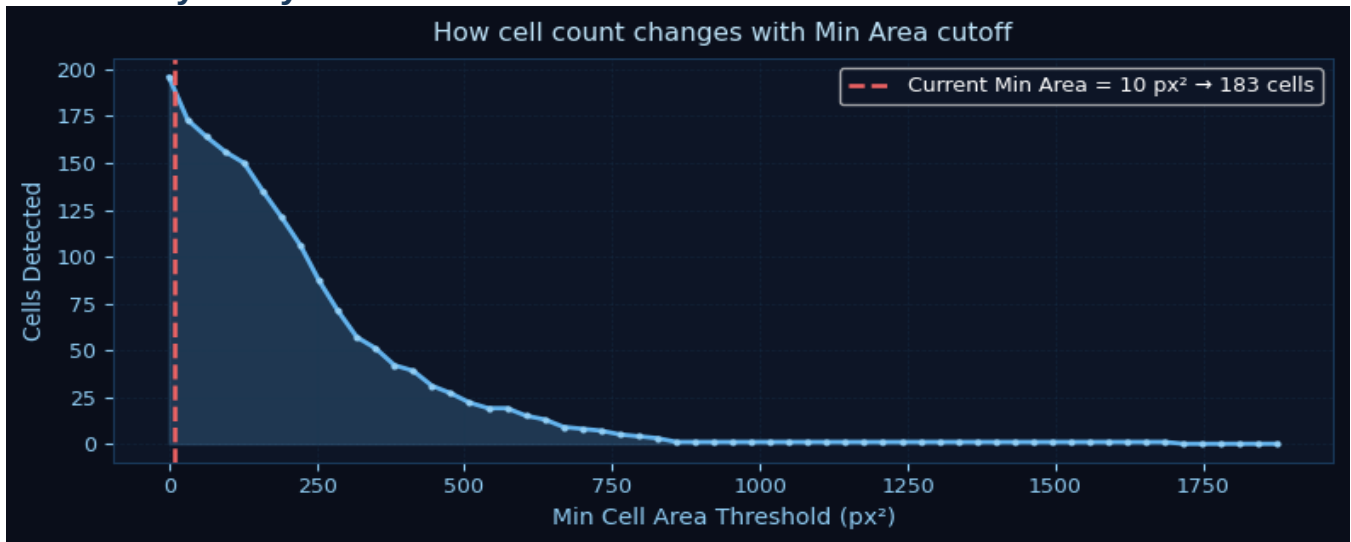
Region of Interest

Detected Cells Overlay

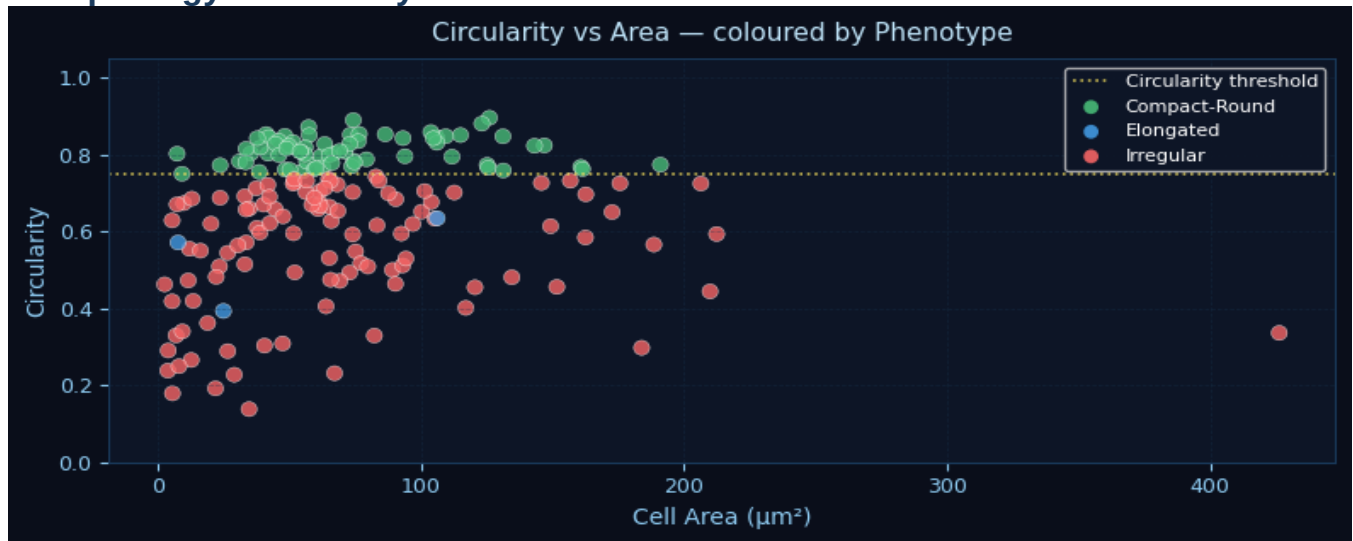
Histogram (Area Distribution)



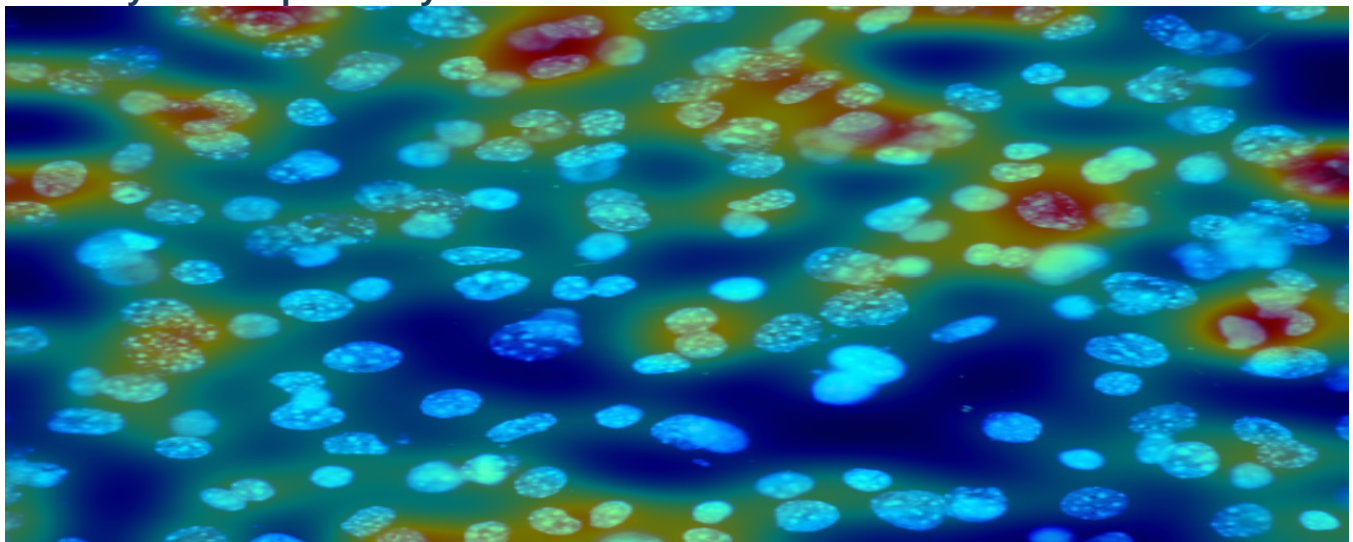
Sensitivity Analysis



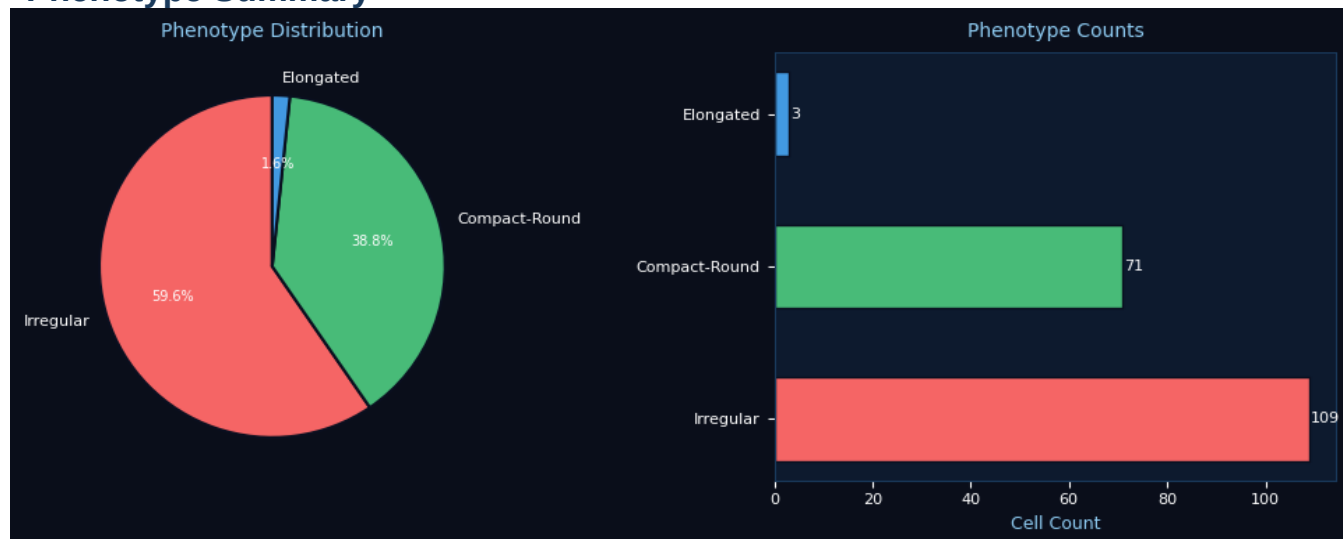
Morphology: Circularity vs Area



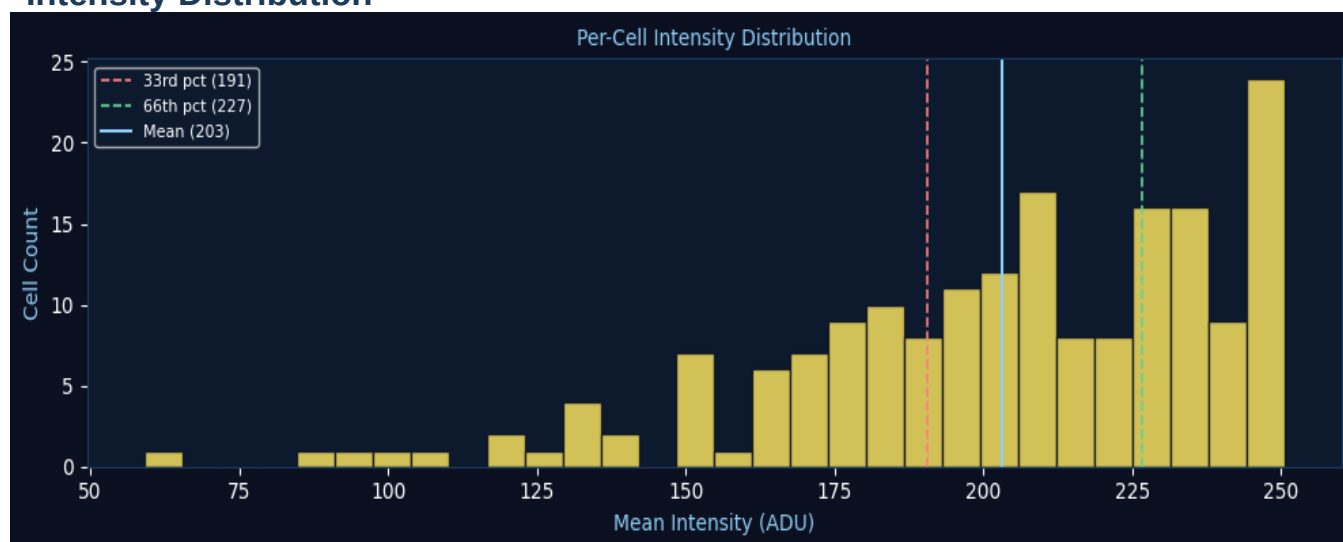
Density Heatmap Overlay



Phenotype Summary



Intensity Distribution



Cell Data Table (first 50 cells)

Cell ID	Area (px²)	Centroid X	Centroid Y	Perimeter	Circularity	Aspect Ratio	Solidity	Mean Intensity	Phenotype	Area (µm²)
1	141.0	266	503	46.97	0.8031	1.2308	0.9724	234.04	Compact-Round	35.25
2	132.0	150	504	48.97	0.6917	1.3077	0.9531	197.01	Irregular	33.0
3	207.0	72	503	59.8	0.7274	1.4	0.9606	140.29	Irregular	51.75
4	87.5	182	502	75.7	0.1919	0.7368	0.4499	230.39	Irregular	21.875
5	21.5	2	488	25.41	0.4183	0.25	1.0	117.74	Irregular	5.375
6	46.0	226	488	34.97	0.4727	0.7692	0.7419	245.23	Irregular	11.5
7	124.0	499	488	44.63	0.7824	0.8125	0.9502	250.48	Compact-Round	31.0
8	247.5	208	489	62.53	0.7955	0.7391	0.9465	226.59	Compact-Round	61.875
9	245.5	138	488	66.18	0.7043	0.6538	0.9497	205.89	Irregular	61.375
10	423.5	470	488	79.36	0.8451	0.6897	0.9769	159.92	Compact-Round	105.875
11	415.0	304	483	77.94	0.8585	0.7778	0.9685	230.72	Compact-Round	103.75
12	154.0	268	479	50.63	0.755	0.6842	0.9277	249.5	Compact-Round	38.5
13	94.5	231	477	39.21	0.7723	0.8571	0.931	246.04	Compact-Round	23.625
14	643.0	344	477	102.57	0.7681	0.75	0.9428	209.27	Compact-Round	160.75
15	167.5	82	467	51.21	0.8025	0.7778	0.9599	236.93	Compact-Round	41.875
16	460.0	414	472	82.43	0.8508	0.7931	0.9654	173.18	Compact-Round	115.0
17	401.0	370	472	87.94	0.6516	0.5806	0.9021	227.48	Irregular	100.25
18	139.0	94	458	51.46	0.6597	0.6842	0.877	230.29	Irregular	34.75
19	297.0	296	454	64.77	0.8897	0.9091	0.977	227.11	Compact-Round	74.25
20	193.0	328	449	53.46	0.8487	0.7895	0.9674	248.39	Compact-Round	48.25
21	298.0	203	451	69.11	0.784	0.5926	0.9707	208.63	Compact-Round	74.5
22	164.5	124	446	49.21	0.8535	0.7222	0.9763	248.36	Compact-Round	41.125
23	193.5	473	446	56.53	0.761	0.6087	0.9532	229.55	Compact-Round	48.375
24	375.5	163	447	77.01	0.7956	0.75	0.9482	240.48	Compact-Round	93.875
25	331.5	10	444	74.87	0.7432	0.8696	0.9312	199.99	Irregular	82.875
26	243.5	182	439	68.18	0.6582	0.5357	0.9154	250.05	Irregular	60.875
27	168.5	409	430	50.04	0.8456	0.8824	0.9601	249.78	Compact-Round	42.125
28	106.0	491	429	49.46	0.5446	0.8235	0.8249	166.92	Irregular	26.5
29	229.0	67	419	57.46	0.8717	1.0	0.9662	198.73	Compact-Round	57.25
30	690.5	446	427	115.5	0.6505	0.725	0.891	227.52	Irregular	172.625

Cell ID	Area (px²)	Centroid X	Centroid Y	Perimeter	Circularity	Aspect Ratio	Solidity	Mean Intensity	Phenotype	Area (µm²)
31	267.0	132	415	64.63	0.8033	1.1667	0.9674	207.55	Compact-Round	66.75
32	468.0	480	409	121.05	0.4013	0.5532	0.7112	174.63	Irregular	117.0
33	764.0	264	405	111.4	0.7737	0.9714	0.9386	242.64	Compact-Round	191.0
34	263.5	196	399	72.67	0.627	1.0	0.9246	223.15	Irregular	65.875
35	372.5	383	399	74.53	0.8428	0.7692	0.9688	245.85	Compact-Round	93.125
36	250.5	71	396	64.04	0.7675	0.6957	0.9312	197.22	Compact-Round	62.625
37	437.5	30	394	80.53	0.8478	1.1304	0.9615	196.09	Compact-Round	109.375
38	415.5	441	392	87.84	0.6767	0.7667	0.8822	161.57	Irregular	103.875
39	183.5	233	389	52.53	0.8358	0.75	0.9683	249.54	Compact-Round	45.875
40	345.5	169	376	71.36	0.8527	0.8	0.9732	239.06	Compact-Round	86.375
41	597.0	114	383	110.57	0.6137	0.641	0.8444	234.99	Irregular	149.25
42	178.0	50	367	58.28	0.6585	1.25	0.8967	230.12	Irregular	44.5
43	205.5	423	358	55.7	0.8324	0.8	0.9536	239.08	Compact-Round	51.375
44	134.0	110	354	50.63	0.657	1.5	0.8508	225.31	Irregular	33.5
45	329.0	37	355	112.08	0.3291	1.4	0.6438	183.98	Irregular	82.25
46	606.5	319	359	129.3	0.4559	0.7209	0.7483	249.87	Irregular	151.625
47	198.0	249	339	55.11	0.8192	1.1176	0.9565	245.55	Compact-Round	49.5
48	190.0	66	336	61.11	0.6393	0.8947	0.8501	177.95	Irregular	47.5
49	587.5	487	338	94.67	0.8238	1.0345	0.9537	220.59	Compact-Round	146.875
50	387.5	329	335	88.67	0.6194	0.8966	0.8767	250.12	Irregular	96.875